

SRAYAN BHATTACHARYA

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EDUCATION

Institute of Engineering and Management, Kolkata, WB

Jul. 2023 – Jul 2027

Bachelor of Technology in Computer Science & Information Technology

CGPA: 8.59/10 (Through Semester 5)

TECHNICAL SKILLS

Languages: Java, Python, SQL(PostgreSQL), JavaScript, C

Frameworks & Libraries: Spring Boot, React, Node.js, TensorFlow, PyTorch, Keras

Developer Tools: Git, Docker, WordPress

Domains: Machine Learning, IoT Systems, Full Stack Development

PROJECTS

ReTAQMLN²: Real-Time Air Quality Monitoring Using Liquid Neural Networks

- Designed and deployed an IoT-based air quality monitoring system using Raspberry Pi 5 integrated with SDS011 and BMP280 sensors for real-time PM_{2.5}, PM₁₀, temperature, pressure, and humidity measurement
- Engineered and trained a Liquid Neural Network (LNN) for short-term pollutant forecasting, achieving R² scores of 0.90 (PM_{2.5}) and 0.94 (PM₁₀), outperforming LSTM and GRU baselines across mid- and long-term horizons
- Applied multivariate time-series preprocessing including z-score normalization, cyclic temporal encoding, delta transformation, and sliding window sequence construction over a 10,000-sample dataset
- Built an interactive React dashboard integrated with Firebase for real-time visualization of pollution metrics and AI-driven forecasts
- Technologies Used:** Python, Liquid Neural Networks, PyTorch, React, Firebase, JavaScript

CodeDuel: Real-Time Multiplayer Competitive Coding Platform

- Architected and deployed a full-stack multiplayer coding platform supporting real-time room management, live code execution, and competitive scoring across concurrent sessions
- Implemented WebSocket-based real-time communication using STOMP protocol for live participant updates, round synchronization, and leaderboard broadcasting
- Integrated Piston API for sandboxed multi-language code execution (Python, Java, C/C++) and Gemini LLM for AI-generated coding challenges with automated test case generation
- Secured the application with JWT-based stateless authentication and role-based access control, and deployed across a multi-cloud infrastructure (GCP, Render, Vercel)
- Technologies Used:** Java, Spring Boot, Spring Security, WebSocket, PostgreSQL, React, TypeScript, Tailwind CSS, Docker

EXPERIENCE

Freelance Web Developer

Jul 2025 – Sep 2025

Remote

- Designed and developed a client-focused e-commerce platform for Creative India Store using WordPress and Elementor
- Translated business requirements into functional website components and storefront navigation workflows
- Built responsive UI ensuring cross-device accessibility and improved user engagement
- Optimized performance and SEO fundamentals to enhance usability and discoverability

RELEVANT COURSEWORK

- Data Structures and Algorithms
- Database Management Systems
- Machine Learning Fundamentals
- Software Engineering